1) Project Title

AgriSense – Design and Development of an AI-Based Soil Health Analysis and Crop Advisory Web System

2) Statement About the Problem

Modern agriculture increasingly relies on soil sensing devices that generate data related to soil nutrients such as Nitrogen (N), Phosphorus (P), Potassium (K), pH value, moisture, and other parameters. While these devices provide accurate measurements, the raw data is often complex and difficult for farmers to interpret without technical expertise.
As a result, farmers struggle to make informed decisions regarding crop selection, fertilizer usage, and soil treatment, leading to reduced productivity and inefficient resource utilization.
There is a strong need for an intelligent software system that can analyze soil data and provide simple, actionable crop advisory recommendations.

3) Why Is This Particular Topic Chosen

This topic is chosen because:

- Agriculture is a backbone of the Indian economy
- Farmers need **decision support**, not raw numerical data
- AI can bridge the gap between soil data and farming decisions
- The project aligns with **Smart Agriculture and Digital India initiatives**
- It offers real-world impact using software-based intelligence

4) Objective and Scope

Objectives

- To analyze soil health data using AI techniques
- To classify soil quality and nutrient levels
- To recommend suitable crops based on soil parameters

- To provide fertilizer and soil improvement suggestions
- To develop a user-friendly web-based advisory system

Scope

- Applicable for small and large-scale farming
- Can support multiple crop categories
- Works with manual data entry or uploaded soil test data
- Can be extended to weather-based advisory in the future
- Suitable for integration with agricultural portals

5) Methodology

1. Collection of soil parameter data (NPK, pH, moisture, etc.)
2. Data preprocessing and normalization
3. Feature extraction from soil data
4. AI-based soil classification and crop recommendation
5. Rule-based advisory generation for fertilizers
6. Backend processing and database storage
7. Web interface development for farmer interaction
8. Testing and validation using sample datasets

6) Hardware & Software Requirements

Hardware Requirements

- Computer or Laptop
- Minimum 8 GB RAM
- Internet connectivity
- Soil data source (manual entry or dataset)

Software Requirements

- Frontend: HTML, CSS, JavaScript, Bootstrap
- Backend: Python (Django / Flask)
- Database: MySQL / SQLite
- AI/ML: Python, Pandas, NumPy, Scikit-learn
- Tools: VS Code, Web Browser
- OS: Windows / Linux

7) Contributions of the Project to Society

- Helps farmers make data-driven agricultural decisions
- Improves crop yield and soil sustainability
- Reduces misuse of fertilizers and chemicals
- Supports eco-friendly and precision farming
- Enhances digital literacy among farmers
- Contributes to food security and rural development

8) Schedule of the Project

Phase	Activity
Week 1	Problem study & literature review
Week 2	Requirement analysis & system planning
Week 3	Data collection & preprocessing
Week 4	AI model design
Week 5–6	AI implementation & testing
Week 7	Backend development
Week 8	Frontend development
Week 9	System integration & validation
Week 10	Documentation & final submission

Task / Week	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Problem Study & Literature Review	✓	✓								
Requirement Analysis & System Planning		✓								
Soil Data Collection & Preprocessing			✓	✓						
AI Model Design & Soil Classification				✓	✓					
Crop Recommendation & Advisory Logic					✓	✓				
Backend Development & Database						✓	✓			
Frontend Development (Farmer UI)							✓			
System Integration & Testing								✓		
Documentation & Final Submission								✓	✓	

9) Limitations of the Project

- Accuracy depends on quality of soil data
- Recommendations are advisory, not a replacement for experts
- Limited crop coverage in initial version
- Does not directly control agricultural hardware
- Requires periodic updates for best results

10) References

1. FAO – Soil Health and Sustainable Agriculture Reports
2. Government of India – Digital Agriculture Initiatives
3. Scikit-learn Documentation
4. Research papers on AI in Precision Farming
5. Python and Django Official Documentation